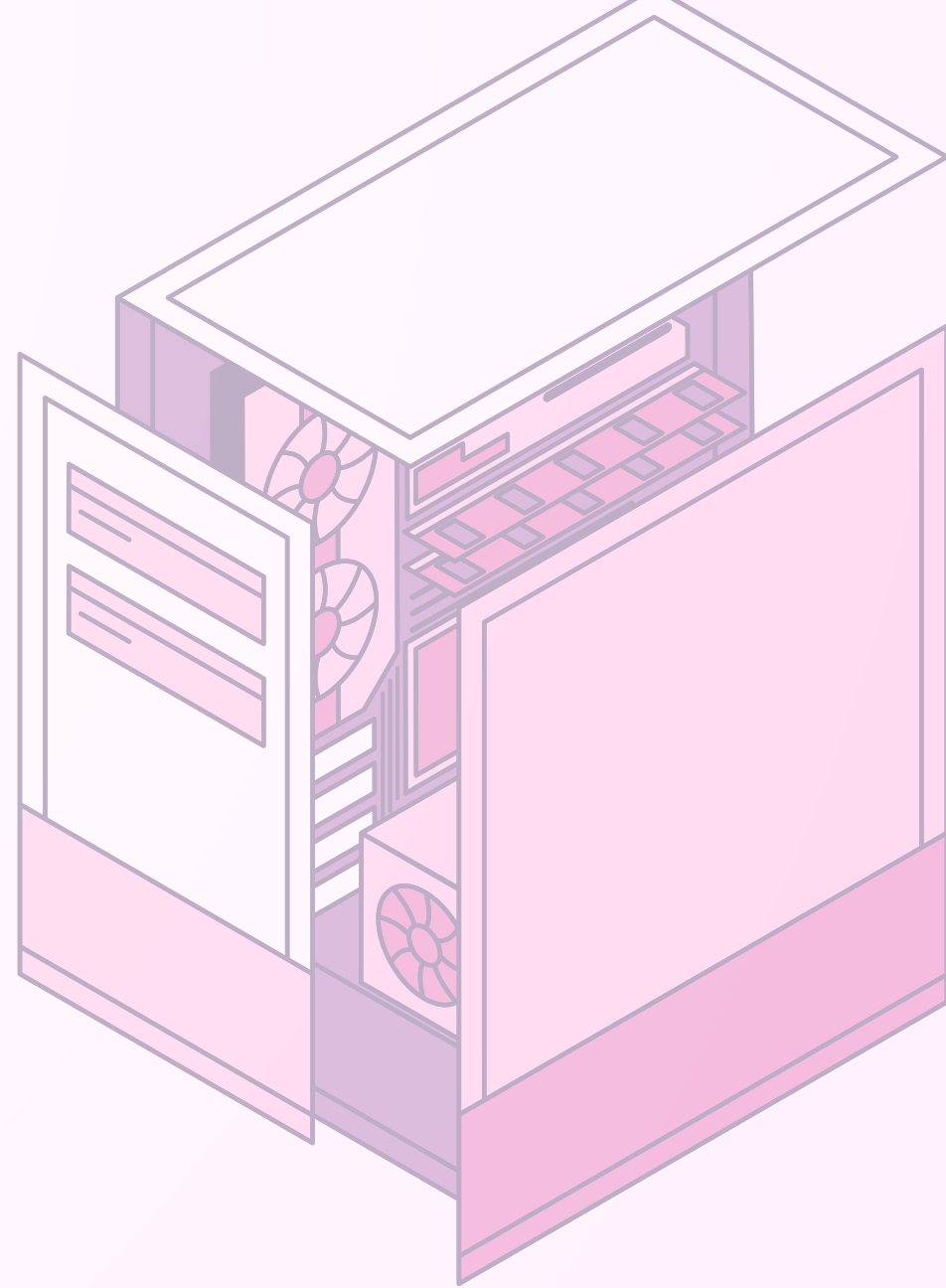




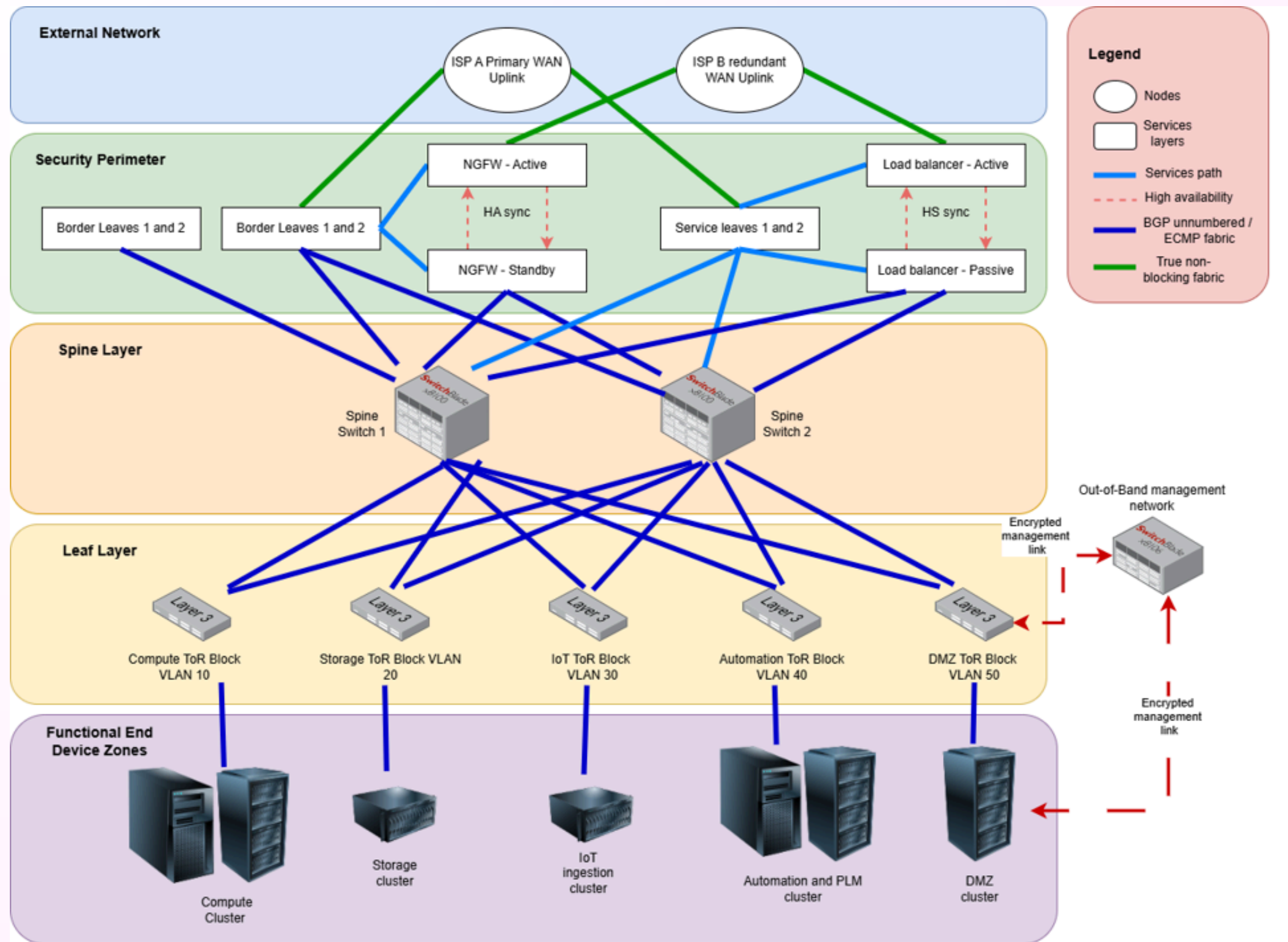
DATA CENTER INFRASTRUCTURE



SYAHMI DANIAL TP077396



Network Topology



Layer 1: WAN/Internet

- Dual ISP (A=primary, B=backup) via separate Border Leaf switches
- BGP auto-failover between ISPs

Layer 2: Spine Layer

- 2× modular Spine chassis (MDA) – ECMP load-balanced
- 350+ x100GbE ports, BGP Unnumbered routing
- 10× EoR racks house passive ODFs only

Layer 4: Smart Alert & Explainable AI

- 350× ToR Leaf, dual 100GbE uplinks to both Spines
- 5 zones grouped logically (not physically): **Compute/Storage/IoT/Automation-PLM/DMZ**
- Distributed gateway, routing happens at first hop, not central point

Layer 3: Security Perimeter

- 2× NGFW (Active/Standby) via Service Leaf
- Zone-based deny-all policy, App-ID enforcement

Layer 5: Storage Leaf

- Dual-fabric: 25GbE Ethernet (NAS) + 32G FC (block) : same rack, separate protocols

Layer 6: End Device Zones

- Compute (VMs), Storage (all-flash SAN/NAS), IoT (Kafka/Spark), Automation/PLM (SCADA, isolated via VXLAN), DMZ (reverse proxy/API gateway)

OOB Management

- Physically separate stackable switch network, VPN-only access, works even if production fabric down

IT HARDWARE

Spine Layer switch



Cisco Nexus 9508

1. VS Arista 7508R3

- **Throughput & Density Deficit**

432 ports and 115 Tbps vs Cisco's 576 100GbE ports and 172.8 Tbps; risks core fabric congestion under smart-factory volumes

2.VS Juniper QFX10008

- **Bandwidth Limitation**

Only ~96 Tbps of backplane bandwidth (nearly half of Cisco's 172.8 Tbps), creating processing bottlenecks.

Higher Latency

3-5 microsecond latency compared to sub-5 microsecond thresholds on dense loads.

Leaf Layer switch



Arista 7050SX3-48YC12

1. VS Cisco Nexus 93180YC-FX3

- **Switching Delay Bottleneck**

~1 microsecond store-and-forward architecture vs Arista's ultra-low 800ns cut-through forwarding technology

- **Buffer Constraints**

Segmented 25MB buffer vs Arista's fully shared 32MB dynamic

2.VS Juniper QFX5120-48Y

- **Port & Uplink Deficit**

8x 100GbE uplinks vs Arista's 12x 100GbE QSFP layout, restricting LACP active-active designs

- **Lower Throughput**

4.0 Tbps cap vs Arista's full 4.8 Tbps

Border Router



Juniper MX204

1.VS Cisco ASR 1001-HX

- **Critical Throughput Shortfall**

60 Gbps capability vs Juniper's 400 Gbps. Risks throttling global factory WAN connectivity paths.

- **Scale Limitation**

Fixed interface boundaries vs Juniper's native 4x 100GbE + 8x 10GbE density

2.VS Fortinet FortiGate 3700D

- **BGP Memory Exhaustion**

UTM-focused appliance with limited RAM capacity; unable to fully hold carrier-grade multi-ISP eBGP routing maps

IT HARDWARE

Compute Zone Server



VS HPE ProLiant DL380 Gen11

1. VS Arista 7508R3

- **Thermal & Density Efficiency Deficit**

Relies on traditional 2.5" drive formats instead of high-density, thermally optimized Gen5 EDSFF E3.S NVMe layouts

- **Hardware Acceleration Ceiling**

Limits scalability to 3 GPUs vs Dell's capacity for up to 6 single-wide processing cards

2. VS Lenovo ThinkSystem SR650 V3

- **Compute Core Deficit**

Caps out at 60 cores per socket (120 total) vs Dell's 64 cores per socket (128 total); reduces VM orchestration density

Storage Array



NetApp AFF A900

1. VS Lenovo ThinkSystem DM7200F

- **Scale-Up Overhead**

Requires purchasing external drive shelves to expand raw capacity vs NetApp's native high-capacity chassis.

- **Cache & IOPS Constraints**

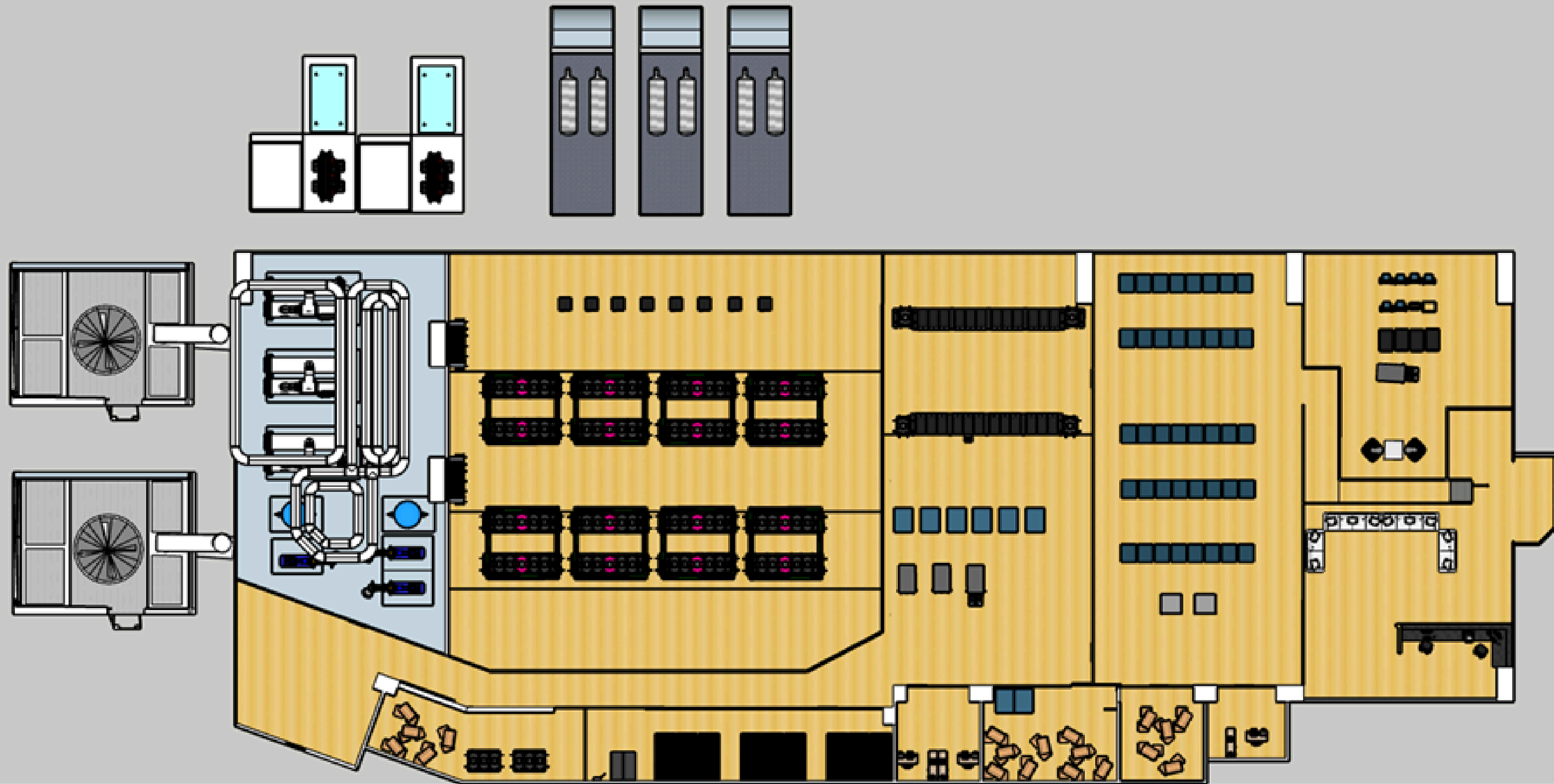
256GB memory environment cannot comfortably match NetApp's 2.4 Million IOPS floor.

2. VS Dell PowerStore 9200T

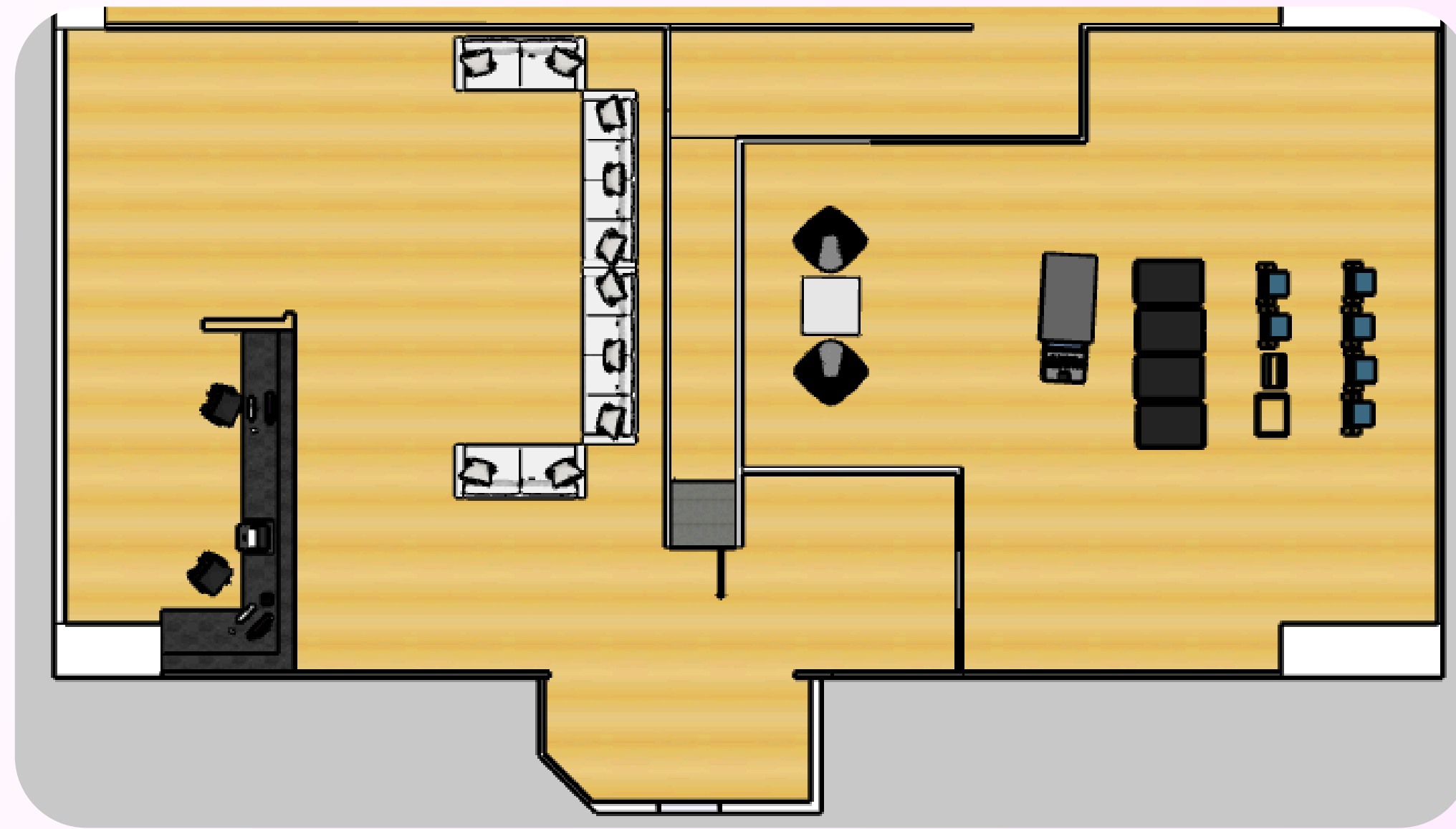
- **Fabric Capacity Ceiling**

Caps capacity at 5.9 PB per appliance and 93 total SSD positions vs NetApp's massive 14.7 PB threshold and 480 NVMe paths.

Data center floor design proposal



Entrance Room and Lobby



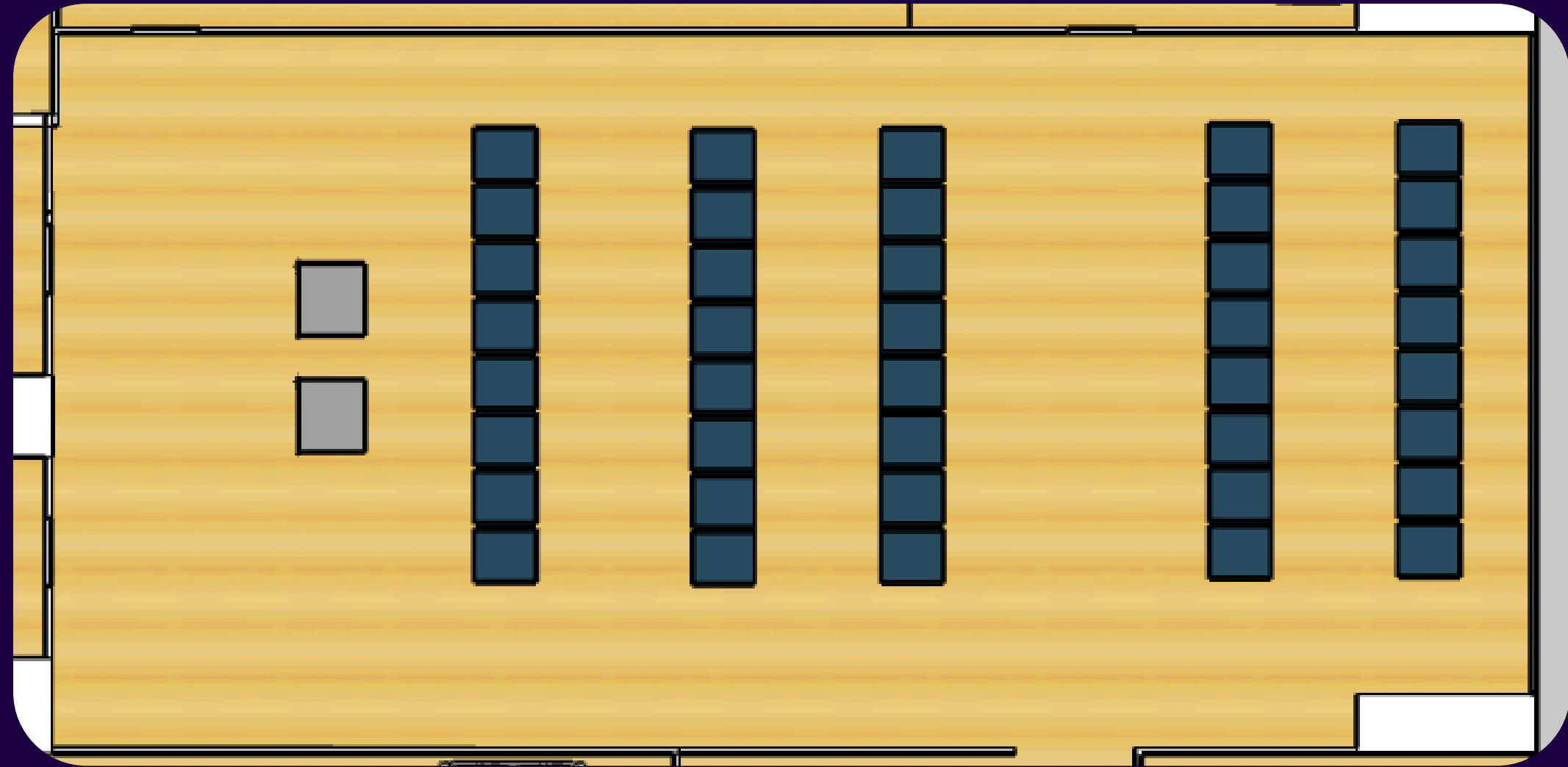
Entrance Room (ER)

- Acts as the perimeter fiber entry zone separating third-party telecom infrastructure lines from internal networks.
- Houses main patch panels and optical terminals to secure external multi-ISP dynamic path feeds.

Lobby

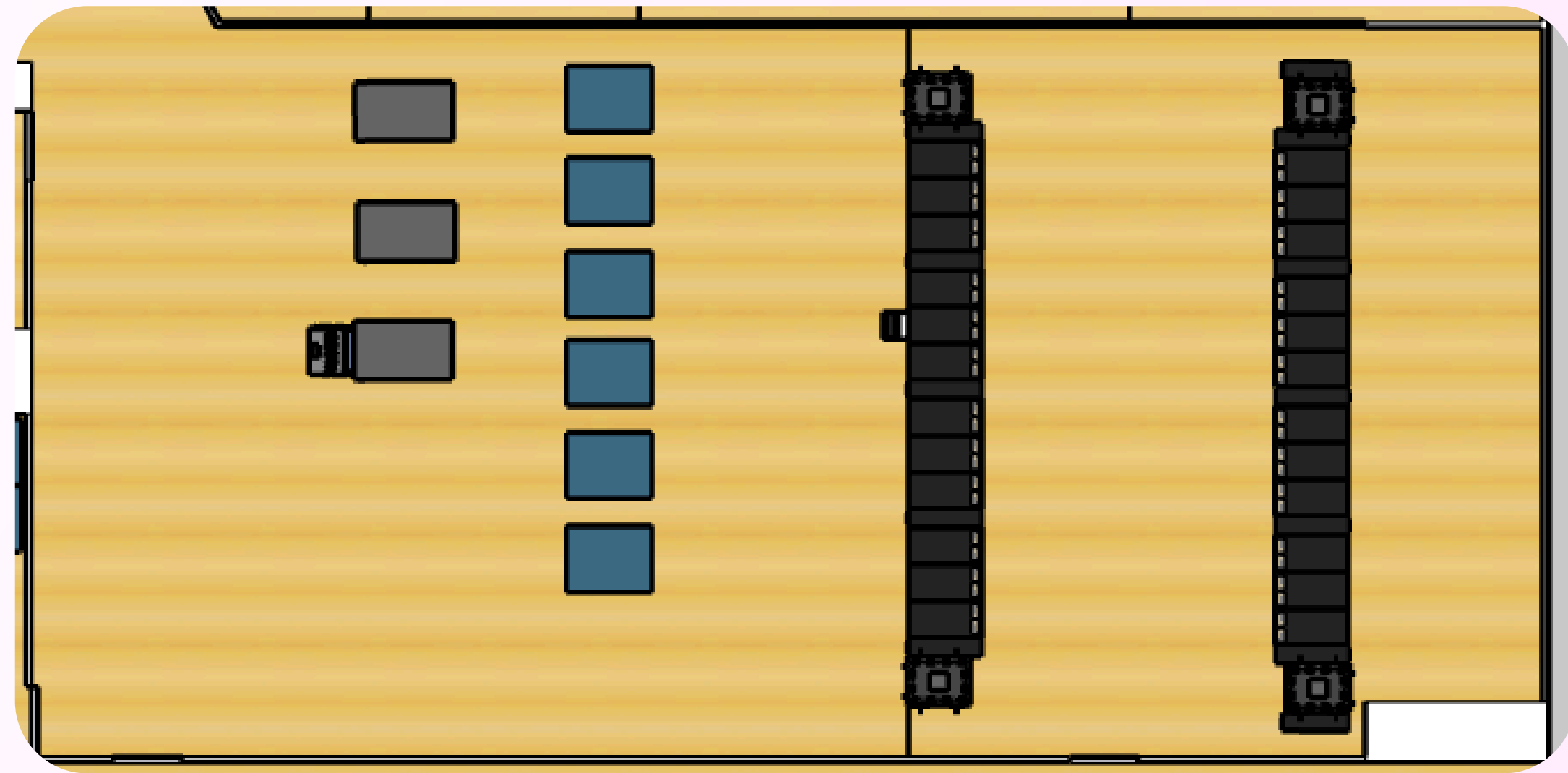
- Administrative entry boundary featuring an L-shaped guest desk to filter unauthorized site visitors.
- Incorporates a dual-door biometric/RFID interlocking mantrap corridor preventing physical tailgating.

Horizontal Distribution Area (HDA)



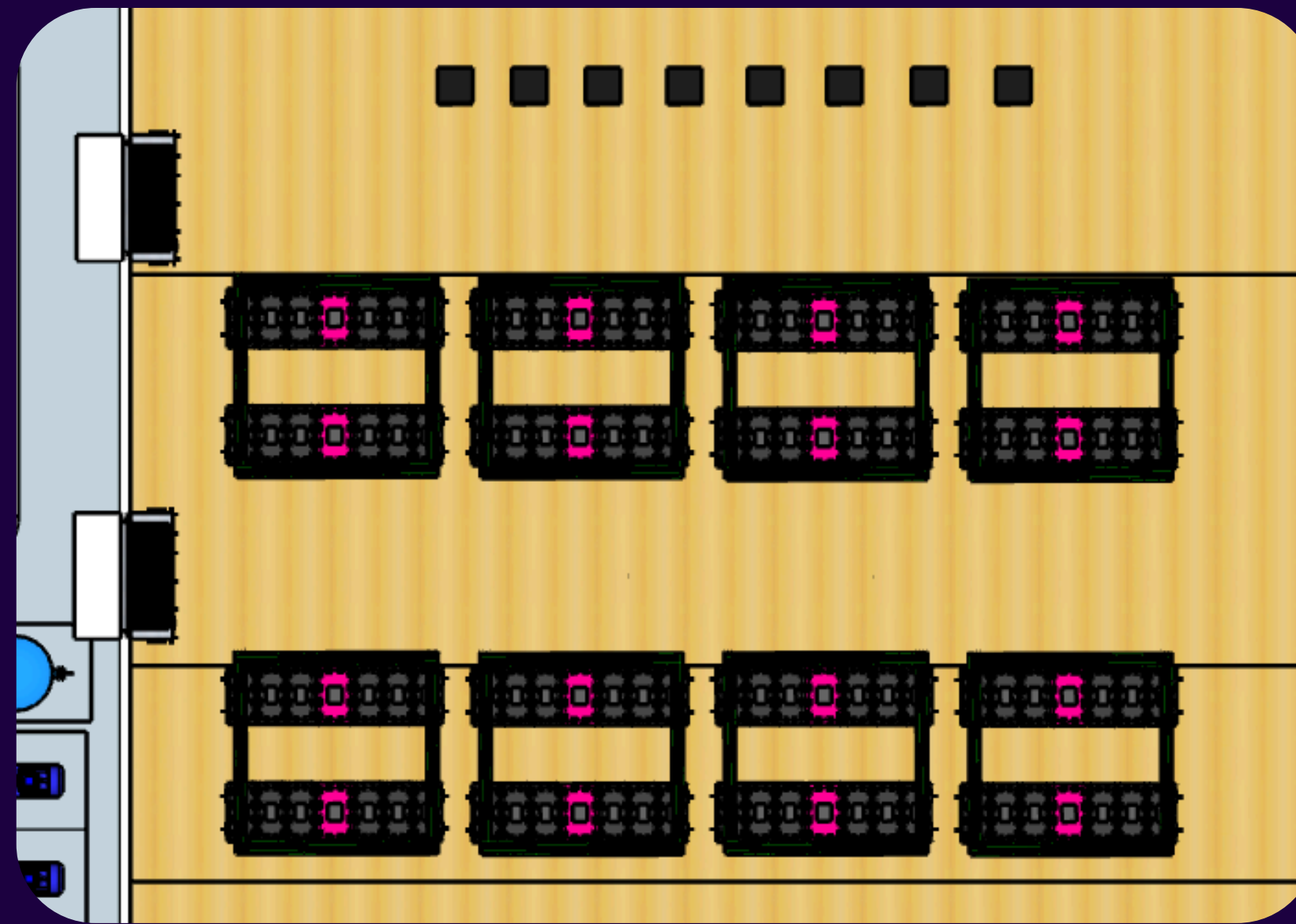
- Functions as the mid-level distribution and structural backbone aggregation tier.
- Employs 5 parallel rows of full-height network cabinets hosting passive cross-connect fields to streamline cabling runs.

Main Distribution Area (MDA)



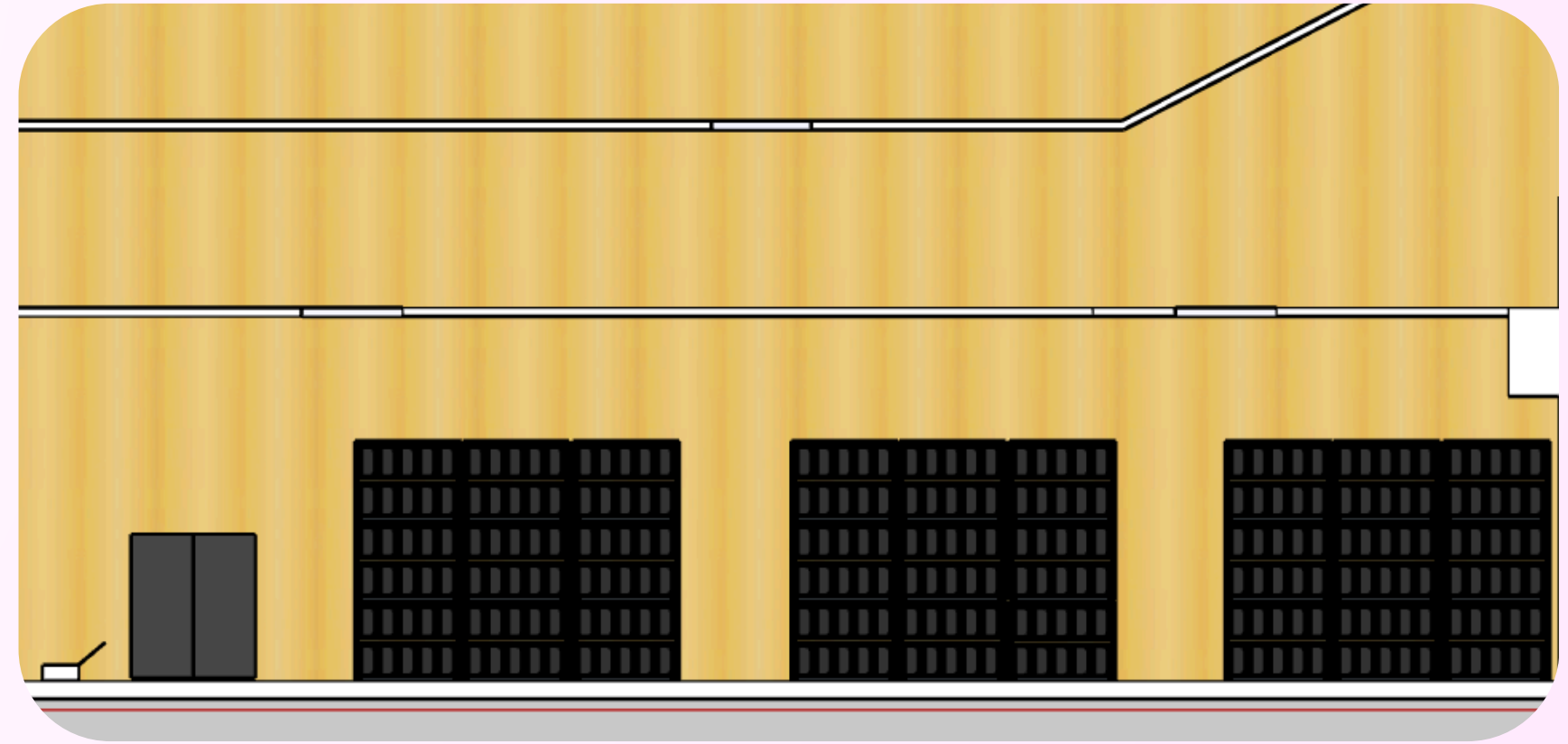
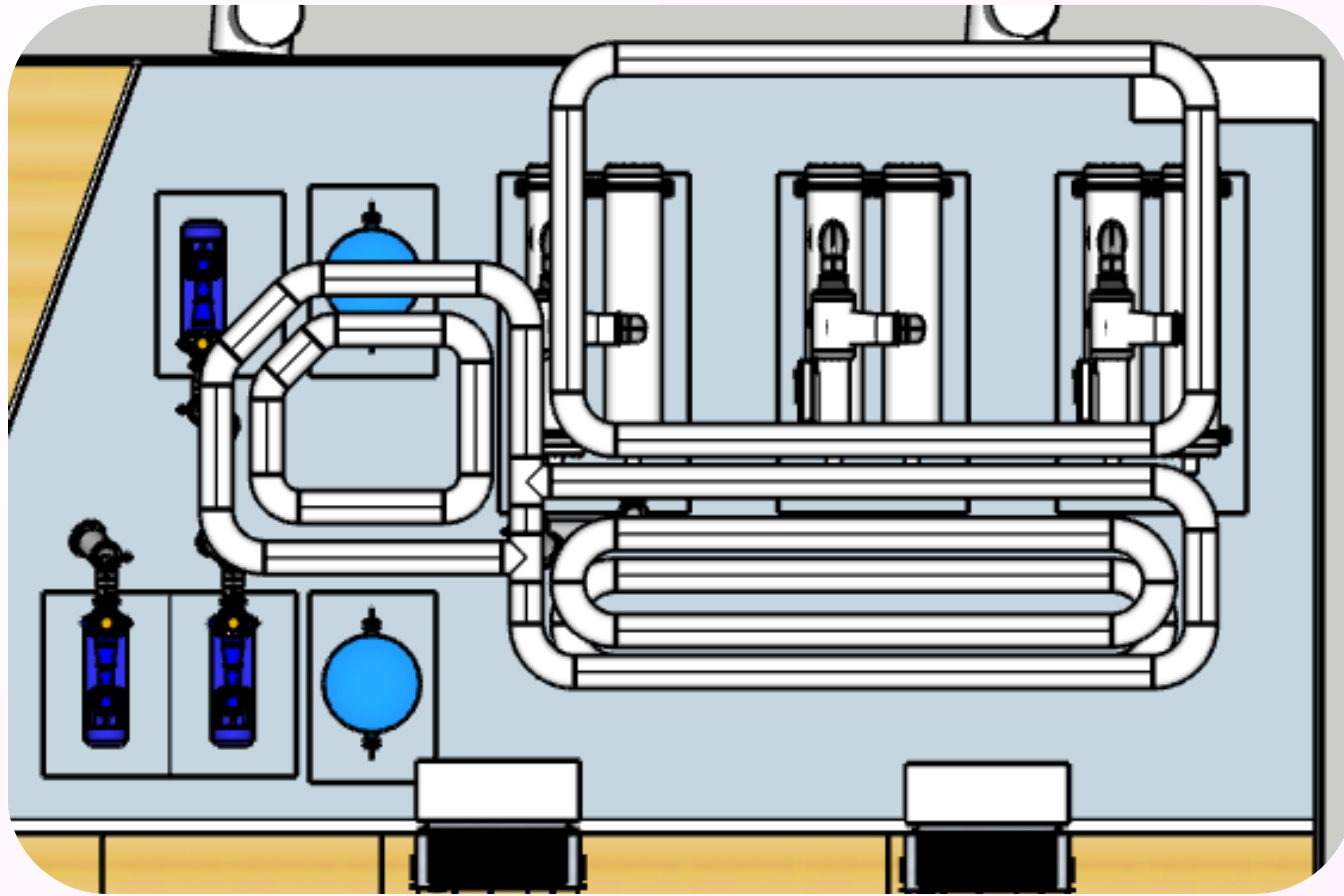
- The primary network engine containing core Cisco Nexus 9508 spine switches in a loop-free fabric.
- Centrally positioned to ensure minimal backbone optical line runs directly to adjacent distribution zones.

Equipment Distribution Area (EDA)

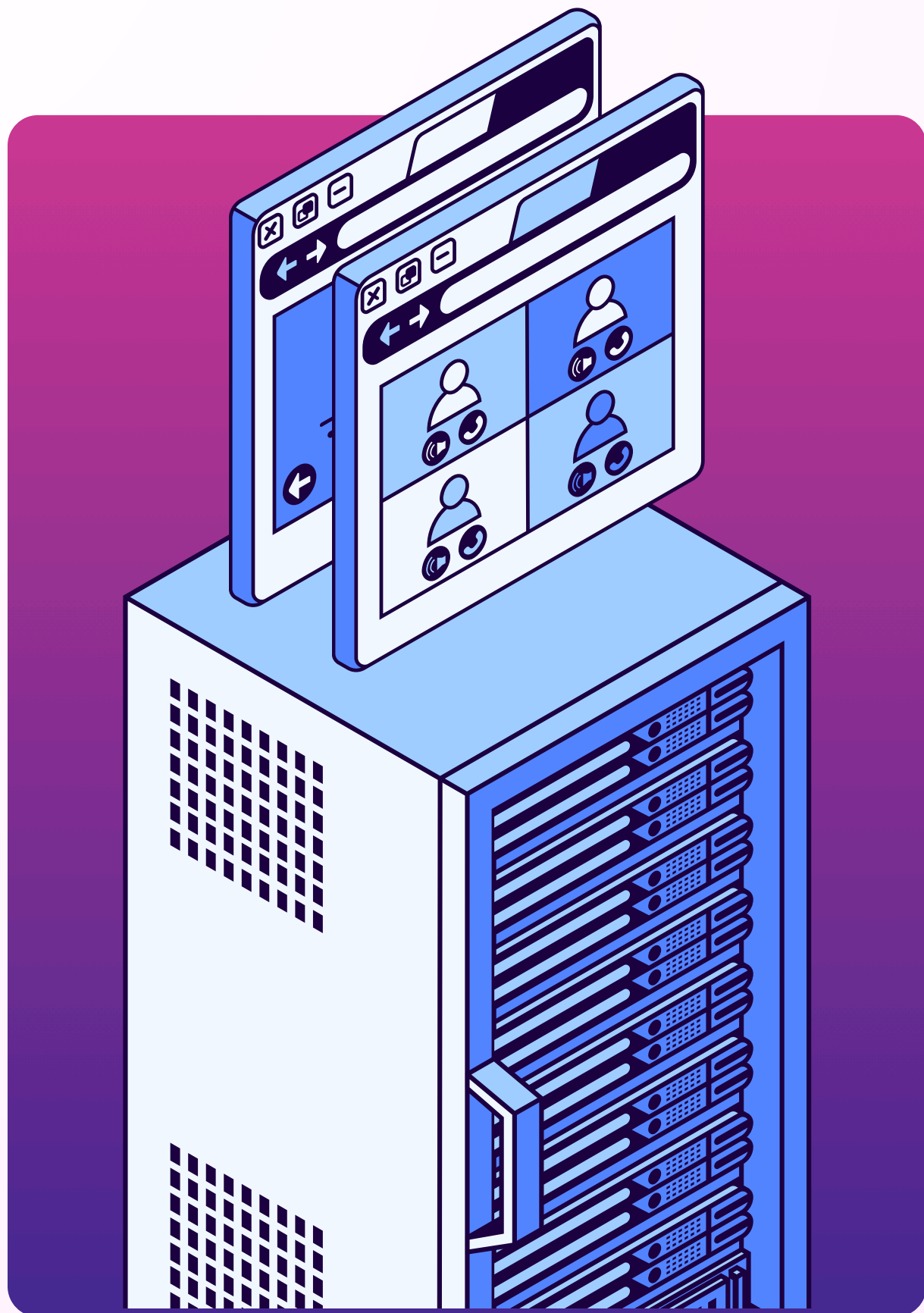


- The core data floor holding 8 rows of sealed active enclosures configured in strict Hot/Cold aisle matrices.
- Hosts the production workloads: 350 Arista Leaf units, Dell R760 nodes, and the NetApp AFF storage tier.

Battery & Cooling Mechanical Utility Rooms



- Battery Room: Parallel UPS arrays and 3 VRLA/Lithium battery rack banks enforcing Tier III power maintenance.
- Cooling Room: Liquid cooling loop infrastructure featuring plumbing manifolds, dual pumps, and twin wall CRAH coils.



THANK YOU